

left

0 ka game every

`int[] arr = {3, 2, 1, 6, 5, 4, 2}; int n = arr.length;`
`int[] left_max = new int[n];`

`left_max[0] = arr[0];`

`for (int i = 1; i < n; i++) {`

`left_max[i] = max(left_max[i-1], arr[i]);`

`}`

already there.
`left_max = [3, 0, 0, 0, 0, 0, 2]`

`left_max`
`[3, 0, 0, 0, 0, 0, 2]`
`i = 0`
`i = 1`
`i = 2`

1 → jo current, wo aaga, wli store bda hai.

2 → store bda na jo chalta aata hai

Right

`arr = [3, 2, 1, 6, 5, 4, 2]`
`int n = arr.length;`

`int[] right_max = new int[n];`

handled

`right_max[n-1] = arr[n-1];`

`for (int i = n-2; i >= 0; i--) {`

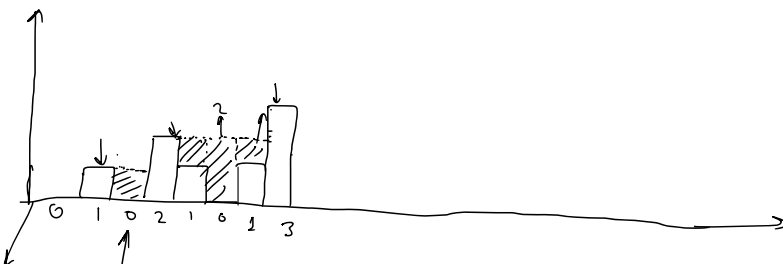
`right_max[i] = max(right_max[i+1], arr[i]);`

`}`

`right_max`
`[0, 0, 0, 0, 0, 0, 2]`
`max(2, 4)`
`max(4, 5)`
`max(5, 6)`
`max(6, 5)`
`max(5, 4)`
`max(4, 3)`

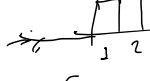
`right_max[i+1]`
`arr[i]`

`[0, 0, 0, 0, 0, 0, 2]`
`n-2`



No water can be trapped

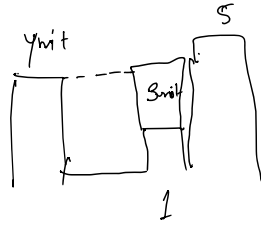
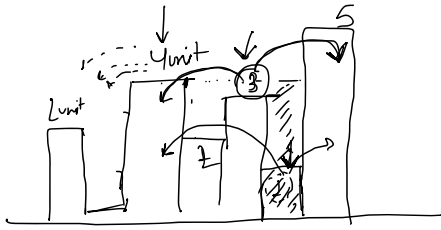
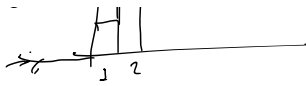
6 unit



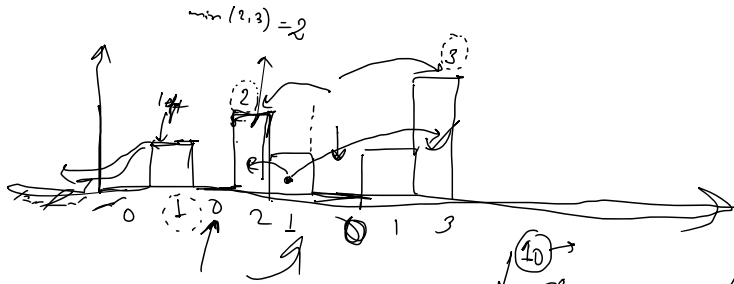
unit



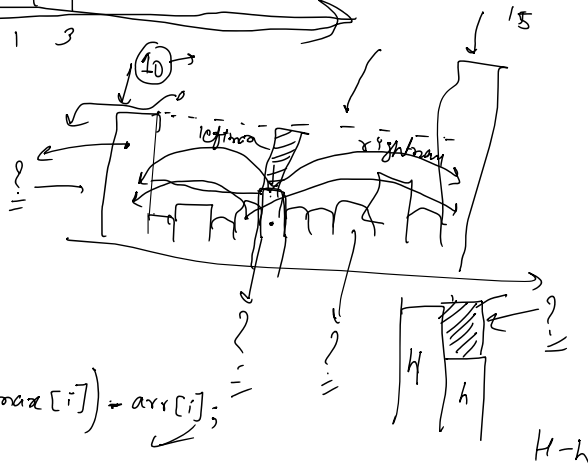
can be trapped



$$\min(4, 5) = 4 \text{ unit} - 1 \text{ unit}$$



$\min(f)$



$$\min(\text{leftmax}[i], \text{rightmax}[i]) - \text{arr}[i];$$

store

$$\text{int ans} = 0;$$

for (int i = 0; i < heights.length; i++)

$$\text{ans} += \min(\text{leftmax}[i], \text{rightmax}[i]) - \text{arr}[i];$$

}

return ans;

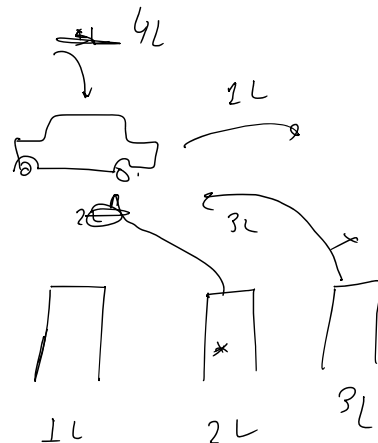
$$\text{gas} = [1, 2, 3, 4, 5]$$

$$\text{cost} = [3, 4, 5, 1, 2]$$

$$C+1 = 7L - 3 = 4L + 2L = 6L - 4L$$

$$= 2L$$

$$= 2L + 3L$$



$$4-1 = 3L + 5L$$

$$= 8L - 2L$$

$$= 6L$$

$$[(1), (2), (3, 4, 5)]$$

$$[3, 4, 5, 1, 2]$$

$$= 2L + 3L$$

$$= 5L - 5L = 0$$

3Ans =



$$5L - 2L = 3L$$

$$3L + 1L = 4L$$

$$4L - 3L = 1L + 2L$$

$$= 3L$$